Implementation of Shrinkage Estimators for Cosmological Precision Matrices

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Abstract. We embed direct precision shrinkage estimators from cosmological statistics into the Ensemble Kalman Filter, recomputing shrinkage coefficients at each assimilation cycle. Six estimators are tested: three direct precision shrinkage variants from Looijmans et al. (identity, eigenvalue, and cosmological targets), a scaled identity variant we propose as a domain-agnostic alternative, the NERCOME estimator, and Ledoit-Wolf covariance shrinkage. Experiments on Lorenz 96 ($n = 40$) and BOSS DR12 cosmological mock data ($d = 18$) show that direct precision shrinkage performance depends on how well the target matches the true precision structure—the methods fail on Lorenz 96 (dense off-diagonal precision) but give the best standalone estimates on near-diagonal cosmological data. Good standalone estimation does not imply good filtering: Ledoit-Wolf produces the worst precision estimates on cosmological data yet the lowest EnKF RMSE, because its low variance pays off when errors compound over repeated assimilation cycles.

Keywords: Data Assimilation · Ensemble Kalman Filter · Shrinkage Estimation · Precision Matrix · Covariance Matrix Estimation

1 Introduction

Ensemble Kalman filters require the inverse of the background error covariance matrix, but in most operational settings the ensemble is far too small to estimate that matrix reliably. When the state dimension n exceeds the ensemble size N_e, the sample covariance is rank-deficient and cannot be inverted at all [10]; even when $N_e > n$, sampling noise introduces spurious correlations that contaminate the precision matrix [1]. The same problem appears in cosmological data analysis, where the number of mock catalogs available to estimate power spectrum covariances is often comparable to the data dimension [8].

Shrinkage estimation addresses this by combining the noisy sample estimate with a structured, low-variance target matrix, accepting some bias in exchange for reduced mean squared error [12]. Most shrinkage work operates in covariance space: Ledoit and Wolf [12] derived an optimal intensity for shrinking toward a scaled identity, Pope and Szapudi [17] brought this idea to cosmology, and Chen et al. [4] proposed oracle-approximating algorithms that improve on Ledoit-Wolf

in high dimensions. Bodnar et al. [3] took a different route, shrinking the precision matrix directly rather than first shrinking the covariance and then inverting. Looijmans et al. [13] applied this direct precision framework to cosmological data, comparing several target choices and showing that domain-informed targets outperform generic ones when the true precision is near-diagonal.

Two gaps remain. First, Looijmans et al. used these estimators for standalone precision matrix estimation, not inside a sequential filter. It is unclear whether an estimator that performs well in a one-shot setting will also perform well when applied repeatedly across hundreds of assimilation cycles, where errors in the precision estimate feed back through the forecast model. Second, the direct precision targets in the literature are either generic (identity, eigenvalues) or domain-specific (cosmological), with no middle ground for practitioners who lack prior knowledge of the precision structure but want something better than the identity.

We address both gaps. We embed the Bodnar et al. direct precision shrinkage framework inside the dual formulation of the Ensemble Kalman Filter, recomputing the shrinkage coefficients at each assimilation cycle. We propose a scaled identity target that uses empirical per-variable variances as a domain-agnostic alternative to the cosmological target. And we test all methods—three targets from Looijmans et al., our scaled variant, NERCOME [11], and Ledoit-Wolf—on two systems with opposite precision structure: Lorenz 96 [14] ($n = 40$, dense off-diagonal precision) and BOSS DR12 cosmological mock data [13] ($d = 18$, near-diagonal precision). The experiments show that direct precision shrinkage performance depends on how well the target matches the true precision structure, and that the estimator producing the worst standalone precision estimates (Ledoit-Wolf) yields the best EnKF filtering error, because its low variance pays off over repeated assimilation cycles. All implementations are publicly available in the TEDA framework [16].

2 Preliminaries

2.1 Data Assimilation

Data assimilation combines a numerical model forecast with observational data to produce a corrected state estimate. Given a state vector $\mathbf{x} \in \mathbb{R}^{n\times 1}$ and observations $\mathbf{y} \in \mathbb{R}^{m\times 1}$ related to the state through an observation operator $\mathbf{H} \in \mathbb{R}^{m\times n}$, the analysis step corrects the forecast by weighting the model and observations according to their respective uncertainties. The background error covariance $\mathbf{P}^b \in \mathbb{R}^{n\times n}$ and the observation error covariance $\mathbf{R} \in \mathbb{R}^{m\times m}$ determine how much to trust each source of information.

2.2 Ensemble Kalman Filter

The Ensemble Kalman Filter (EnKF) [5] represents the state using an ensemble of N_e state vectors $\{\mathbf{x}_i^b\}_{i=1}^{N_e}$, each $\mathbf{x}_i^b \in \mathbb{R}^{n\times 1}$. The ensemble mean

$$\bar{\mathbf{x}}^b = \frac{1}{N_e}\sum_{i=1}^{N_e}\mathbf{x}_i^b \in \mathbb{R}^{n\times 1}, \tag{1}$$

serves as the background state estimate. With the deviation matrix

$$\Delta\mathbf{X}^b = \mathbf{X}^b - \bar{\mathbf{x}}^b\mathbf{1}^T \in \mathbb{R}^{n\times N_e}, \tag{2}$$

where $\mathbf{X}^b = [\mathbf{x}_1^b, \ldots, \mathbf{x}_{N_e}^b]$, the sample covariance is

$$\mathbf{P}^b = \frac{1}{N_e - 1}\Delta\mathbf{X}^b(\Delta\mathbf{X}^b)^T \in \mathbb{R}^{n\times n}. \tag{3}$$

The stochastic EnKF computes analysis updates by solving

$$[\mathbf{H}\mathbf{P}^b\mathbf{H}^T + \mathbf{R}]\,\Delta\mathbf{X}^a = \mathbf{D}^s \in \mathbb{R}^{m\times N_e}, \tag{4}$$

where $\mathbf{D}^s$ is the innovation matrix of perturbed observation departures.

The dual EnKF [15] reformulates this in terms of the precision matrix:

$$[(\mathbf{P}^b)^{-1} + \mathbf{H}^T\mathbf{R}^{-1}\mathbf{H}]\,\Delta\mathbf{X}^a = \mathbf{H}^T\mathbf{R}^{-1}\mathbf{D}^s \in \mathbb{R}^{n\times N_e}. \tag{5}$$

This formulation requires the precision matrix $(\mathbf{P}^b)^{-1}$—the central object of this work. When $N_e < n$, the sample covariance (3) has rank at most $N_e - 1$ and cannot be inverted. Even when $N_e > n$, the sample precision can be poorly conditioned, amplifying noise in the analysis.

A third variant projects the analysis onto the ensemble subspace [16]:

$$\mathbf{X}^a = \mathbf{X}^b + \Delta\mathbf{X}^b\mathbf{W}^*, \tag{6}$$

where $\mathbf{W}^* \in \mathbb{R}^{N_e\times N_e}$ is obtained from a reduced system, bringing the problem from n dimensions down to N_e.

2.3 Covariance Matrix Estimation

Shrinkage estimation blends the sample covariance with a structured target $\mathbf{T}$:

$$\hat{\mathbf{P}} = (1-\lambda)\mathbf{S} + \lambda\mathbf{T}, \tag{7}$$

where $\lambda \in [0,1]$ controls how much weight goes to the target. The optimal λ minimizes

$$\lambda^* = \arg\min_{\lambda\in[0,1]} \mathbb{E}\left[\|\hat{\mathbf{P}} - \boldsymbol{\Sigma}\|_F^2\right], \tag{8}$$

with $\boldsymbol{\Sigma}$ the true covariance.

Ledoit-Wolf estimator. Ledoit and Wolf [12] derived a closed-form optimal intensity for the target $\mathbf{T} = \mu\mathbf{I}$, $\mu = \mathrm{tr}(\mathbf{S})/n$:

$$\lambda^*_{\mathrm{LW}} = \frac{\sum_{i=1}^{N_e} \|\mathbf{z}_i\mathbf{z}_i^T - \mathbf{S}\|_F^2 / N_e^2}{\|\mathbf{S} - \mu\mathbf{I}\|_F^2}, \tag{9}$$

where $\mathbf{z}_i$ are standardized observations. This estimator is asymptotically optimal and has become a standard reference in high-dimensional covariance estimation.

Oracle estimator. When the true covariance $\boldsymbol{\Sigma}$ is known (as in simulation studies), the oracle intensity

$$\lambda^*_{\mathrm{oracle}} = \frac{\mathrm{tr}[(\mathbf{S} - \boldsymbol{\Sigma})(\mathbf{T} - \boldsymbol{\Sigma})]}{\|\mathbf{T} - \mathbf{S}\|_F^2} \tag{10}$$

gives a best-case benchmark.

NERCOME estimator. NERCOME [11] sidesteps the choice of target entirely. It randomly splits the ensemble into two disjoint subsets, diagonalizes one subset's sample covariance via eigendecomposition, rotates the other subset's covariance into that eigenbasis keeping only the diagonal, and averages over many such random splits. The split ratio that minimizes cross-validated loss is selected, producing a better-conditioned estimate without structural assumptions about the covariance.

Pope and Szapudi [17] introduced shrinkage estimation to cosmology, showing that even a simple diagonal target improves power spectrum covariance estimation when few simulations are available. Looijmans et al. [13] later compared covariance shrinkage against direct precision shrinkage (following Bodnar et al. [3]) and NERCOME, finding that incorporating cosmological prior information helps, especially at larger sample sizes.

2.4 Inverse Covariance Matrix Estimation

Rather than shrinking the covariance and then inverting, Bodnar et al. [3] proposed shrinking the precision matrix directly. The estimator combines the sample precision $\hat{\boldsymbol{\Pi}}_S = \mathbf{S}^{-1}$ with a structured target $\boldsymbol{\Pi}_0$:

$$\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0, \tag{11}$$

where α^* and β^* are estimated analytically:

$$\alpha^* = 1 - \frac{n}{N_e} - \frac{N_e^{-1} \|\hat{\boldsymbol{\Pi}}_S\|_{\mathrm{tr}}^2 \, \|\boldsymbol{\Pi}_0\|_F^2}{\|\hat{\boldsymbol{\Pi}}_S\|_F^2 \, \|\boldsymbol{\Pi}_0\|_F^2 - \big(\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)\big)^2}, \tag{12}$$

$$\beta^* = \left(1 - \frac{n}{N_e} - \alpha^*\right) \frac{\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)}{\|\boldsymbol{\Pi}_0\|_F^2}, \tag{13}$$

where $\|\cdot\|_{\mathrm{tr}}$ is the trace norm and $\|\cdot\|_F$ the Frobenius norm. These coefficients are derived for the raw inverse $\mathbf{S}^{-1}$; the $(1 - n/N_e)$ term already accounts for the estimation bias, so the Hartlap correction [9] should not be applied beforehand.

In cosmological data analysis, the data vector consists of power spectrum measurements $\hat{C}_\ell$ at different multipoles ℓ. For a Gaussian random field observed over a sky fraction $f_{\rm sky}$, the covariance of power spectrum estimates is [17]:

$$\mathrm{Cov}(\hat{C}_{\ell_i}, \hat{C}_{\ell_j}) = \frac{2}{(2\ell_i + 1) f_{\rm sky}} \left(C_{\ell_i} + \frac{1}{n_g} \right)^2 \delta_{\ell_i \ell_j}, \tag{14}$$

where C_ℓ is the true power spectrum, n_g the galaxy number density, and $(2\ell + 1) f_{\rm sky}$ counts the independent modes N_ℓ at multipole ℓ. The diagonal structure follows from mode independence under the Gaussian assumption. This leads to a cosmological shrinkage target [17,13]:

$$T_{ii}^{(2)} = \frac{2}{N_{\ell_i}} C_{\ell_i}^2, \tag{15}$$

where $N_{\ell_i} = (2\ell_i + 1) f_{\rm sky}$, capturing the leading-order cosmic variance contribution.

2.5 Modified Cholesky Decomposition

The modified Cholesky decomposition [15] factors the precision matrix as

$$(\mathbf{P}^b)^{-1} \approx \mathbf{L}^{-T} \mathbf{D}^{-1} \mathbf{L}^{-1}, \tag{16}$$

where $\mathbf{L}$ is unit lower triangular and $\mathbf{D}$ is diagonal. Each row of $\mathbf{L}$ is estimated by regressing a state variable on its neighbors within a bandwidth r, a form of localization that zeroes out long-range correlations. This produces a sparse, banded precision matrix by construction, avoiding the need to invert $\mathbf{P}^b$ entirely.

2.6 EnKF via Modified Cholesky Decomposition

Substituting (16) into the dual EnKF formulation (5) gives an analysis that uses the banded precision estimate directly. The bandwidth r controls the sparsity: small r enforces locality (reducing noise but potentially discarding real correlations), while large r retains more structure at the cost of higher variance. Nino-Ruiz et al. [15] showed that even modest bandwidths ($r = 2$–5) substantially improve the EnKF on spatially extended systems like Lorenz 96.

3 Proposed Method

3.1 Direct Precision Shrinkage for the EnKF

The Bodnar et al. [3] framework was developed for standalone precision matrix estimation and applied to cosmological data by Looijmans et al. [13]. Neither work embedded the estimator inside a sequential filter. We do so here: at each

Algorithm 1 Direct Precision Shrinkage (Identity Target)

Require: Deviation matrix $\mathbf{DX} \in \mathbb{R}^{n \times N_e}$
1: Compute sample covariance: $\mathbf{S} = \frac{1}{N_e - 1} \mathbf{DX}\,\mathbf{DX}^T$
2: Compute sample precision: $\hat{\boldsymbol{\Pi}}_S = \mathbf{S}^{-1}$
3: Set target: $\boldsymbol{\Pi}_0 = \mathbf{I}_n$
4: Compute α^*, β^* via Eqs. (12)–(13)
5: **return** $\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0$

assimilation cycle of the dual EnKF (5), we replace the raw sample precision $\hat{\boldsymbol{\Pi}}_S = \mathbf{S}^{-1}$ with the shrinkage estimator

$$\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0, \tag{17}$$

where $\boldsymbol{\Pi}_0$ is a structured target and the coefficients α^*, β^* are computed from Eqs. (12)–(13) at every cycle using the current ensemble.

The weight α^* on the sample precision is bounded above by $1 - n/N_e$, reflecting the known positive bias of the Wishart inverse. The remaining weight β^* on the target depends on how well $\boldsymbol{\Pi}_0$ aligns with $\hat{\boldsymbol{\Pi}}_S$: when $\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)$ is large relative to $\|\boldsymbol{\Pi}_0\|_F^2$, more weight goes to the target. Both coefficients are clamped to $[0, 1]$ with $\alpha^* + \beta^* \leq 1$ as a finite-sample safeguard, since the asymptotic guarantees of Bodnar et al. do not hold at the ensemble sizes typical of data assimilation ($N_e = 20$–100).

Embedding this estimator in the EnKF introduces a consideration absent from standalone estimation: the precision matrix is recomputed at every assimilation cycle, and errors in the analysis state propagate through the forecast model into subsequent cycles. An estimator that is accurate on average but variable from cycle to cycle can accumulate larger errors over a long assimilation window than a slightly biased but stable one. This sequential accumulation effect is not captured by one-shot metrics like Frobenius loss and must be evaluated through filtering experiments (Section 4).

3.2 Shrinkage Targets

The four target variants below differ only in their choice of $\boldsymbol{\Pi}_0$. The first three follow Looijmans et al. [13]; the fourth is a domain-agnostic adaptation we propose.

Identity Target. The simplest choice: $\boldsymbol{\Pi}_0 = \mathbf{I}_n$, assuming independent unit-variance priors. This carries no domain-specific information, but the direct precision formulation still produces a valid precision matrix—unlike inverting a shrunk covariance, which can amplify estimation error.

Eigenvalue Target. A diagonal matrix of the sorted eigenvalues of $\hat{\boldsymbol{\Pi}}_S$: $\boldsymbol{\Pi}_0^{(1)} = \mathrm{diag}(e_1, \ldots, e_n)$ with $e_1 \leq \cdots \leq e_n$. This retains the spectral information of

Algorithm 2 Direct Precision Shrinkage (Eigenvalue Target)

Require: Deviation matrix $\mathbf{DX} \in \mathbb{R}^{n \times N_e}$
1: Compute $\mathbf{S}$, $\hat{\boldsymbol{\Pi}}_S$ as in Algorithm 1 (Steps 1–2)
2: Compute eigenvalues $e_1 \leq \cdots \leq e_n$ of $\hat{\boldsymbol{\Pi}}_S$
3: Set target: $\boldsymbol{\Pi}_0^{(1)} = \mathrm{diag}(e_1, \ldots, e_n)$
4: Compute α^*, β^* via Eqs. (12)–(13)
5: **return** $\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0^{(1)}$

Algorithm 3 Direct Precision Shrinkage (Cosmological Target)

Require: Deviation matrix $\mathbf{DX}$, power spectrum C_ℓ, mode counts N_ℓ
1: Compute $\mathbf{S}$, $\hat{\boldsymbol{\Pi}}_S$ as in Algorithm 1 (Steps 1–2)
2: **for** each data element i mapped to multipole ℓ_i **do**
3: Compute diagonal covariance target: $T_{ii}^{(2)} = \frac{2}{N_{\ell_i}} C_{\ell_i}^2$
4: **end for**
5: Set target precision: $\boldsymbol{\Pi}_0^{(2)} = (\mathbf{T}^{(2)})^{-1}$
6: Compute α^*, β^* via Eqs. (12)–(13)
7: **return** $\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0^{(2)}$

the sample precision while zeroing out off-diagonal correlations, and requires no external domain knowledge. It corresponds to the first precision target ($\boldsymbol{\Pi}_0^{(1)}$) in Looijmans et al. [13].

Cosmological Target. The analytical target from Looijmans et al., derived from the Gaussian power spectrum covariance (14). Each diagonal entry of the target covariance is $T_{ii}^{(2)} = (2/N_{\ell_i}) C_{\ell_i}^2$, encoding cosmic variance at multipole ℓ_i [8], and the target precision is its inverse: $\boldsymbol{\Pi}_0^{(2)} = (\mathbf{T}^{(2)})^{-1}$. When the assumed power spectrum model is close to the truth, this provides a stronger shrinkage direction than the generic alternatives.

Scaled Identity Target (Novel Adaptation). The cosmological target requires power spectrum data C_ℓ and mode counts N_ℓ that are unavailable outside cosmology. We propose a domain-agnostic alternative that uses empirical per-variable variances from the sample covariance:

$$\boldsymbol{\Pi}_0 = \mathrm{diag}\left(\frac{1}{2\sigma_1^2}, \ldots, \frac{1}{2\sigma_n^2} \right), \tag{18}$$

where σ_i^2 are the diagonal entries of $\mathbf{S}$. The factor of 2 mirrors the $2/N_\ell$ scaling of the cosmological target (15). This variant tests how much of the direct precision shrinkage performance depends on domain-specific target knowledge versus the general framework itself.

Table 1. Computational complexity per assimilation cycle. n: state dimension, N_e: ensemble size, r: Cholesky bandwidth, B: number of NERCOME bootstrap draws.

Method	Cost per cycle
EnKF (standard)	$O(n^2 N_e + m^2 N_e)$
EnKF-Cholesky	$O(n \cdot r \cdot N_e)$
Identity Shrinkage	$O(n^2 N_e + n^3)$
Scaled Shrinkage	$O(n^2 N_e + n^3)$
Cosmological Shrinkage	$O(n^2 N_e + n^3)$
Eigenvalue Shrinkage	$O(n^2 N_e + 2n^3)$
NERCOME	$O(B \cdot (n^2 N_e + n^3))$
Ledoit-Wolf	$O(n^2 N_e)$

3.3 Computational Complexity

All four direct precision shrinkage variants share a common computational core: forming the sample covariance $\mathbf{S}$ costs $O(n^2 N_e)$, and inverting it to obtain $\hat{\boldsymbol{\Pi}}_S$ costs $O(n^3)$. Computing α^* and β^* requires the Frobenius norm $\|\hat{\boldsymbol{\Pi}}_S\|_F^2$, the trace norm $\|\hat{\boldsymbol{\Pi}}_S\|_{\mathrm{tr}}$, and the product $\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)$, each costing $O(n^2)$ for a dense target or $O(n)$ for a diagonal one. The final linear combination is $O(n^2)$. Since the $O(n^3)$ inversion dominates, all four variants have the same asymptotic cost per assimilation cycle.

They differ in target construction. The identity target requires no computation. The scaled identity reads n diagonal entries of $\mathbf{S}$, costing $O(n)$. The cosmological target similarly costs $O(n)$ given precomputed C_ℓ and N_ℓ values. The eigenvalue target requires a symmetric eigendecomposition of $\hat{\boldsymbol{\Pi}}_S$, costing $O(n^3)$, which doubles the effective cost of that variant.

Table 1 summarizes the per-cycle cost of each method tested in this work, including the baselines.

The modified Cholesky approach avoids the $O(n^3)$ inversion entirely by exploiting banded structure, making it the cheapest option on spatially extended systems where $r \ll n$. Ledoit-Wolf shrinks the covariance directly and does not require inversion ($O(n^2 N_e)$), though the subsequent EnKF solve still involves an $O(m^2 N_e)$ linear system. NERCOME is the most expensive: its B bootstrap resamples (typically $B = 200$) each require a covariance computation and eigendecomposition, resulting in a cost roughly B times that of the other shrinkage methods. On the AR(1) model ($n = 18$, $N_e = 30$), NERCOME takes ∼27 seconds per 100 cycles versus ∼0.2 seconds for the other shrinkage methods. On Lorenz 96 ($n = 40$, $N_e = 80$), the gap widens to ∼215 seconds versus ∼26 seconds.

All implementations are publicly available in TEDA[1].

[1] TEDA source code: https://github.com/enino84/TEDA.git

4 Benchmark

4.1 Experimental Setup

We test on two systems with different precision matrix structures.

Lorenz 96. The Lorenz 96 model [14] has $n = 40$ state variables with forcing $F = 8$, producing chaotic dynamics with dense off-diagonal precision structure due to spatial coupling. We use $m = 32$ randomly placed observations with Gaussian error $\sigma_o = 0.01$, multiplicative inflation $\rho = 1.05$, and 100 assimilation cycles at observation frequency $\Delta t_{\text{obs}} = 0.1$. Ensemble sizes are $N_e \in \{50, 60, 80, 100\}$, all above the $N_e > n + 2 = 42$ threshold required for direct precision methods. Each configuration is repeated 10 times with different random seeds.

AR(1) cosmological model. To test on data with near-diagonal precision structure, we build an AR(1) model calibrated to BOSS DR12 Patchy mock catalog statistics from Looijmans et al. [13]:

$$\mathbf{x}_{t+1} = \phi\, \mathbf{x}_t + (1 - \phi)\, \boldsymbol{\mu} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \tag{19}$$

with $\boldsymbol{\mu}$ the mean power spectrum from the mocks, $\phi = 0.95$, and $\mathbf{Q} = (1 - \phi^2)\boldsymbol{\Sigma}_{\text{mock}}$ so the stationary distribution has covariance $\boldsymbol{\Sigma}_{\text{mock}}$. This preserves the near-diagonal precision structure within a dynamical system. We use $d = 18$ state variables (9 k-bins $\times$ 2 multipoles), $m = 14$ observations ($\approx 78\%$ coverage), $\sigma_o = 100$, inflation $\rho = 1.02$, and $N_e \in \{24, 30, 40, 50\}$.

We also replicate the standalone precision estimation experiment of Looijmans et al. using the same BOSS mock catalogs. For subsample sizes $n \in \{24, 30, 40, 50\}$, we draw $M = 100$ random subsets of n mocks, compute each estimator's precision matrix, and measure the relative Frobenius loss against the true precision (computed from all 2048 mocks with Hartlap correction [9]):

$$\mathcal{L}(\hat{\boldsymbol{\Pi}}, \boldsymbol{\Pi}_{\text{true}}) = \frac{\|\hat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}_{\text{true}}\|_F}{\|\boldsymbol{\Pi}_{\text{true}}\|_F}. \tag{20}$$

Seven methods are compared on Lorenz 96: standard EnKF, EnKF-Cholesky (bandwidth $r = 2$), three direct precision shrinkage variants (identity, scaled identity, eigenvalue targets), NERCOME, and Ledoit-Wolf. The cosmological target is excluded from Lorenz 96 since this model does not produce power spectrum data; it is evaluated on the cosmological experiments alongside all other methods.

For filtering experiments, we report the relative RMSE of the analysis state:

$$\text{RMSE}(t) = \frac{\|\bar{\mathbf{x}}^a(t) - \mathbf{x}^{\text{true}}(t)\|_2}{\|\mathbf{x}^{\text{true}}(t)\|_2}, \tag{21}$$

time-averaged over the last 50 cycles to discard the initial transient.

4.2 Lorenz 96 Results

Figure 1 shows RMSE over 100 assimilation cycles for six methods at $N_e = 100$.

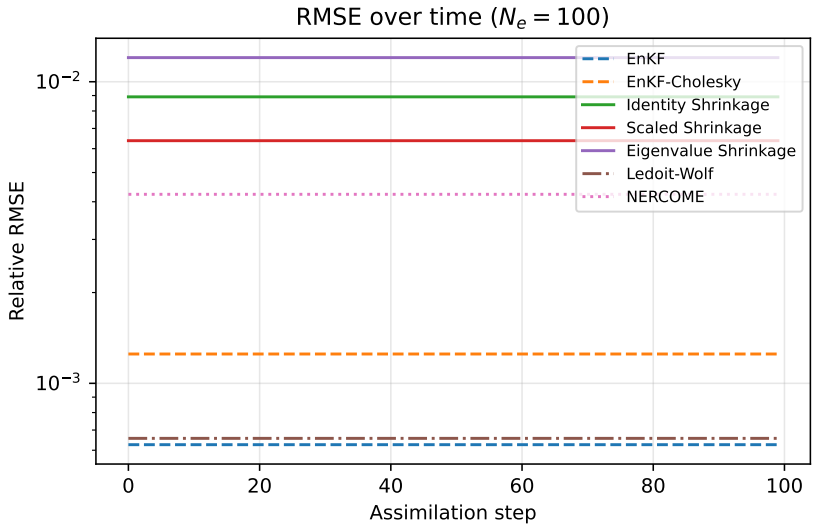

Fig. 1. Evolution of relative RMSE over 100 assimilation cycles for different analysis methods with $N_e = 100$. Shrinkage-based methods (solid lines) and Ledoit-Wolf (dash-dot) are compared against the standard EnKF and EnKF-Cholesky (dashed).

Table 2 summarizes performance across ensemble sizes. Ledoit-Wolf achieves the lowest RMSE at three of four ensemble sizes, with near-zero variance across runs. The direct precision shrinkage methods show erratic non-monotonic behavior—RMSE sometimes jumps at intermediate ensemble sizes before dropping again—and high standard deviations reflecting seed-dependent instability.

This instability traces to the Bodnar coefficients. While α^* stays near $1-n/N_e$ across all ensemble sizes and assimilation steps, β^* is severely miscalibrated: unclamped values reach 10^8–10^9 for the identity target and 10^3–10^5 for the scaled target, requiring clamping to $[0, 1]$ at every step. For the eigenvalue target, β^* is effectively zero ($\sim 10^{-4}$), so no shrinkage toward the target occurs. The identity target $\boldsymbol{\Pi}_0 = \mathbf{I}$ is a poor match for Lorenz 96, whose precision matrix has significant off-diagonal entries from spatial coupling. In the formula for β^* (13), the $\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)/\|\boldsymbol{\Pi}_0\|_F^2$ factor becomes very large when the sample precision has large diagonal entries relative to the target norm, driving β^* to extreme values. After clamping, the resulting precision matrices have condition numbers of 10^7–10^{11}, making the analysis updates numerically unstable.

Figure 2 plots the time-averaged RMSE against N_e. The standard EnKF and Ledoit-Wolf improve monotonically with larger ensembles. Modified Cholesky gives stable though not optimal performance across all sizes.

NERCOME shows a counterintuitive trend: RMSE *increases* with ensemble size (0.23 at $N_e = 50$ to 0.42 at $N_e = 100$). At small N_e, the sample covariance is noisy and bootstrap resampling provides genuine regularization; at larger N_e,

Table 2. Time-averaged relative RMSE ($\times 10^{-2}$) for different methods and ensemble sizes on Lorenz 96. Best result per column in **bold**. †Scaled Shrinkage is a novel domain-agnostic adaptation proposed in this work.

Method	$N_e = 50$	$N_e = 60$	$N_e = 80$	$N_e = 100$
EnKF (standard)	0.35 ± 0.18	0.38 ± 0.56	0.08 ± 0.02	$\mathbf{0.06 \pm 0.01}$
EnKF-Cholesky	0.14 ± 0.01	0.13 ± 0.01	0.13 ± 0.01	0.13 ± 0.01
Identity Shrinkage	0.79 ± 0.62	1.53 ± 1.62	0.63 ± 0.31	0.81 ± 0.82
Scaled Shrinkage†	0.43 ± 0.20	0.88 ± 0.66	0.51 ± 0.24	0.59 ± 0.35
Eigenvalue Shrinkage	1.56 ± 0.59	2.00 ± 2.11	1.68 ± 0.93	1.30 ± 1.25
NERCOME	0.23 ± 0.09	0.29 ± 0.18	0.36 ± 0.15	0.42 ± 0.19
Ledoit-Wolf	$\mathbf{0.08 \pm 0.01}$	$\mathbf{0.08 \pm 0.01}$	$\mathbf{0.07 \pm 0.00}$	0.07 ± 0.01

Table 3. Relative Frobenius loss of precision matrix estimates on BOSS DR12 mock data ($d = 18$). Lower is better. Best result per column in **bold**.

Method	$n = 24$	$n = 30$	$n = 40$	$n = 50$
Sample (Hartlap)	1.15 ± 1.01	0.74 ± 0.44	0.47 ± 0.15	0.43 ± 0.16
Identity Shrinkage	1.18 ± 1.13	0.68 ± 0.43	$\mathbf{0.45 \pm 0.12}$	$\mathbf{0.41 \pm 0.14}$
Scaled Shrinkage†	0.90 ± 0.96	$\mathbf{0.60 \pm 0.36}$	0.51 ± 0.11	0.46 ± 0.13
Eigenvalue Shrinkage	1.20 ± 1.13	0.70 ± 0.43	0.46 ± 0.13	$\mathbf{0.41 \pm 0.14}$
Cosmological Shrinkage	1.19 ± 1.13	0.69 ± 0.43	0.46 ± 0.13	$\mathbf{0.41 \pm 0.14}$
NERCOME	$\mathbf{0.82 \pm 0.06}$	0.73 ± 0.08	0.63 ± 0.08	0.53 ± 0.08
Ledoit-Wolf	0.96 ± 0.01	0.95 ± 0.01	0.94 ± 0.01	0.93 ± 0.01

the sample covariance is already reasonable, and bootstrap variance from 200 draws per cycle compounds over 100 sequential updates.

Figure 3 compares the precision matrices produced by each method at a representative assimilation step.

4.3 Cosmological Results

Standalone Precision Estimation. Table 3 shows the relative Frobenius loss for standalone precision matrix estimation on BOSS DR12 mock data. The results reverse compared to Lorenz 96: the direct precision shrinkage methods now outperform Ledoit-Wolf by a wide margin. At $n = 50$, the identity, eigenvalue, and cosmological targets all achieve losses of ≈ 0.41 versus 0.93 for Ledoit-Wolf. This confirms the findings of Looijmans et al. that direct precision shrinkage works well when the true precision matrix is near-diagonal.

Figure 4 visualizes these trends. Ledoit-Wolf's loss stays nearly flat at ≈ 0.93–0.96 regardless of subsample size: it shrinks the *covariance* toward a diagonal target and then inverts, a suboptimal route when the precision matrix itself is near-diagonal. NERCOME does best at small n thanks to its low variance, and the scaled identity target achieves the lowest mean loss at $n = 30$. Both are

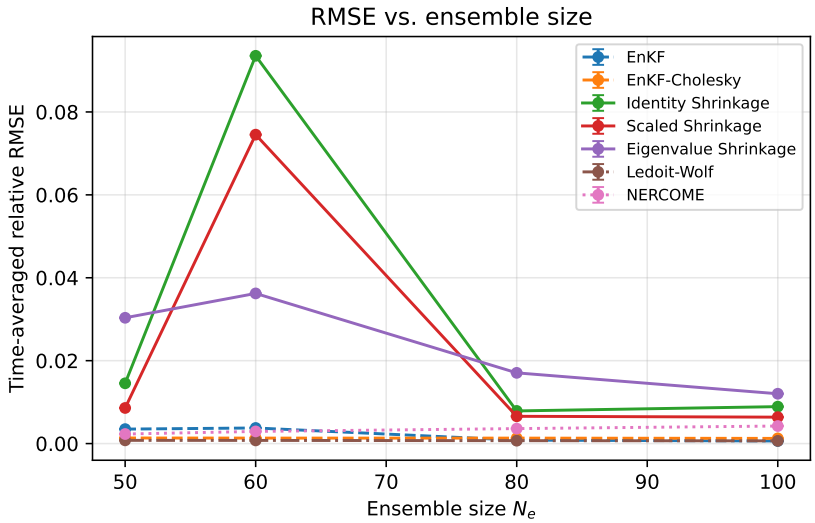

Fig. 2. Time-averaged relative RMSE (over last 50 cycles) as a function of ensemble size N_e on Lorenz 96. Error bars show ± 1 standard deviation over 10 runs.

overtaken by the Bodnar methods once the sample size is large enough for the sample precision to become reliable.

EnKF Filtering. Table 4 reports the time-averaged RMSE for the AR(1) cosmological model. Ledoit-Wolf achieves the lowest RMSE at all four ensemble sizes, despite having the worst standalone precision estimates. Unlike on Lorenz 96, the direct precision shrinkage methods now improve smoothly and monotonically with ensemble size, confirming that the erratic Lorenz 96 behavior was caused by target-truth mismatch, not a fundamental flaw.

Figure 5 plots RMSE against ensemble size. All methods converge as N_e grows. Pairwise Welch's t-tests ($n = 10$ runs) show that on Lorenz 96, Ledoit-Wolf's advantage is statistically significant ($p < 0.05$) at most ensemble sizes. On the cosmological data, most shrinkage methods do not differ significantly from Ledoit-Wolf ($p > 0.05$) at $N_e \geq 40$.

The scaled identity target occupies a middle ground. On Lorenz 96, it outperforms the other direct precision variants (Table 2), probably because scaling by empirical variances gives a better-conditioned target than the plain identity. On cosmological data, it achieves the lowest mean Frobenius loss at $n = 30$ (Table 3) but falls slightly behind the Bodnar methods at larger sample sizes. It serves its intended purpose: modest gains over the identity target without requiring external information, but no substitute for a domain-specific target when one is available.

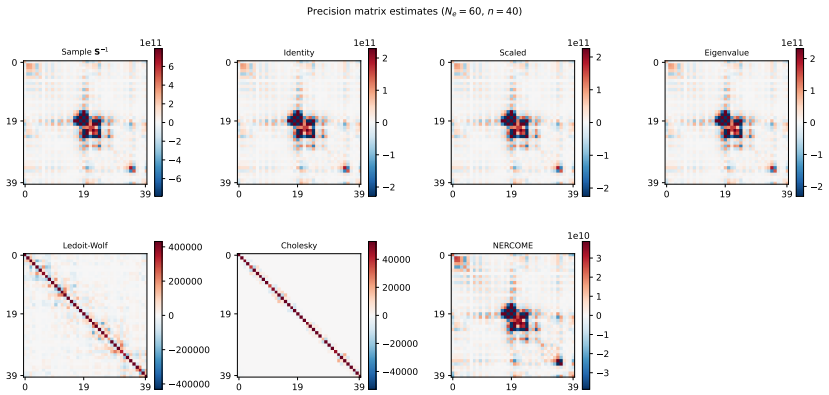

Fig. 3. Precision matrix estimates (40×40) from different methods with $N_e = 60$: sample precision $\mathbf{S}^{-1}$, identity shrinkage, scaled shrinkage, eigenvalue shrinkage, Ledoit-Wolf, and modified Cholesky. The direct precision shrinkage estimators produce matrices with much larger magnitude than Ledoit-Wolf and Cholesky, reflecting the ill-conditioning discussed above.

4.4 Estimation Quality vs. Filtering Performance

The most striking result across both domains is the disconnect between standalone estimation quality and filtering performance. On cosmological data, the Bodnar et al. methods achieve Frobenius losses half those of Ledoit-Wolf (Table 3), yet Ledoit-Wolf yields the lowest EnKF RMSE (Table 4).

The explanation is sequential accumulation. The precision estimator is applied at every assimilation cycle, and step-to-step variability compounds across 100 updates. Ledoit-Wolf's standard deviation across draws is ± 0.01 in Part A—an order of magnitude below the direct precision methods (± 0.13–0.14). A slightly biased but stable estimator beats a more accurate but variable one when errors accumulate over many cycles.

The dual-domain comparison confirms that the poor Lorenz 96 results for direct precision shrinkage reflect target-truth mismatch, not a deficiency of the methods. On Lorenz 96, diagonal targets produce erratic RMSE (0.43–2.00×10^{-2}). On the near-diagonal cosmological data, the same methods achieve stable performance (1.73–2.39×10^{-2}). The Bodnar framework works as designed when the target captures the dominant structure of the true precision matrix.

For sequential data assimilation, variance reduction matters at least as much as bias reduction.

5 Conclusions and Future Work

A good precision matrix estimator and a good filtering estimator are not the same thing. On cosmological data, the Bodnar et al. [3] methods achieve Frobenius losses half those of Ledoit-Wolf, yet Ledoit-Wolf yields the lowest EnKF RMSE.

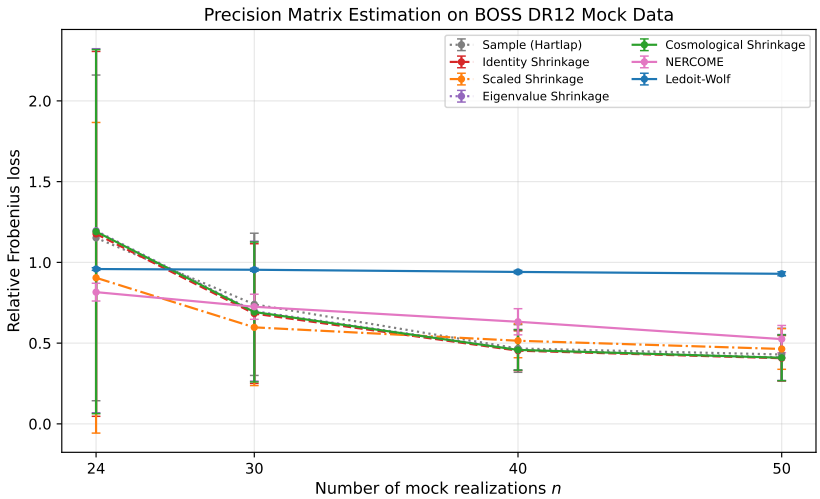

Fig. 4. Relative Frobenius loss of precision matrix estimates vs. subsample size on BOSS DR12 mock data ($d = 18$). Direct precision shrinkage methods (solid lines) achieve the lowest loss at $n \geq 40$, while Ledoit-Wolf (blue) remains flat near 0.93. Error bars show ± 1 standard deviation over 100 random draws.

The reason is stability: a slightly biased estimator that behaves consistently from cycle to cycle accumulates less error over 100 sequential updates than one that is accurate on average but noisy. Without domain-specific prior knowledge, Ledoit-Wolf is the safest default among the methods we tested.

Direct precision shrinkage remains the better choice when the target matches the true precision structure. Our cosmological experiments confirm Looijmans et al. [13]: with near-diagonal data, these methods outperform all alternatives in standalone estimation. Their failure on Lorenz 96 is a consequence of applying diagonal targets to a system whose precision matrix has dense off-diagonal structure, not a deficiency of the framework itself.

The scaled identity target we propose occupies a middle ground—modest gains over the plain identity without requiring domain-specific information, but no substitute for a well-chosen target when one is available. Its ad-hoc factor of 2, borrowed by analogy from the cosmological $2/N_\ell$ term, lacks theoretical justification and could be improved through a data-driven selection criterion.

Several directions remain open. Our implementation does not combine shrinkage with spatial localization [6,15], which could yield further gains on spatially extended systems. The AR(1) model in the cosmological experiments preserves the covariance structure of the BOSS mock catalogs but is far simpler than a realistic cosmological simulation. Future work should test on higher-dimensional systems such as the quasi-geostrophic model, explore adaptive target selection based on observed covariance structure, and investigate whether the sequential

Table 4. Time-averaged relative RMSE ($\times 10^{-2}$) for the AR(1) cosmological model. Best result per column in **bold**. †Scaled Shrinkage is a novel adaptation proposed in this work.

Method	$N_e = 24$	$N_e = 30$	$N_e = 40$	$N_e = 50$
EnKF (standard)	2.93 ± 0.44	2.20 ± 0.34	2.16 ± 0.25	2.06 ± 0.39
EnKF-Cholesky	2.07 ± 0.30	1.76 ± 0.20	1.86 ± 0.27	1.86 ± 0.37
Identity Shrinkage	2.03 ± 0.27	1.74 ± 0.23	1.86 ± 0.21	1.84 ± 0.34
Scaled Shrinkage†	2.04 ± 0.25	1.73 ± 0.20	1.89 ± 0.27	1.85 ± 0.33
Eigenvalue Shrinkage	2.39 ± 0.28	2.00 ± 0.26	2.01 ± 0.23	1.92 ± 0.35
Cosmological Shrinkage	2.25 ± 0.25	2.00 ± 0.28	2.05 ± 0.27	1.97 ± 0.36
NERCOME	1.93 ± 0.23	1.79 ± 0.21	1.88 ± 0.28	1.90 ± 0.33
Ledoit-Wolf	$\mathbf{1.94 \pm 0.27}$	$\mathbf{1.71 \pm 0.22}$	$\mathbf{1.84 \pm 0.22}$	$\mathbf{1.83 \pm 0.34}$

accumulation effect we observed generalizes to other classes of estimators beyond those tested here.

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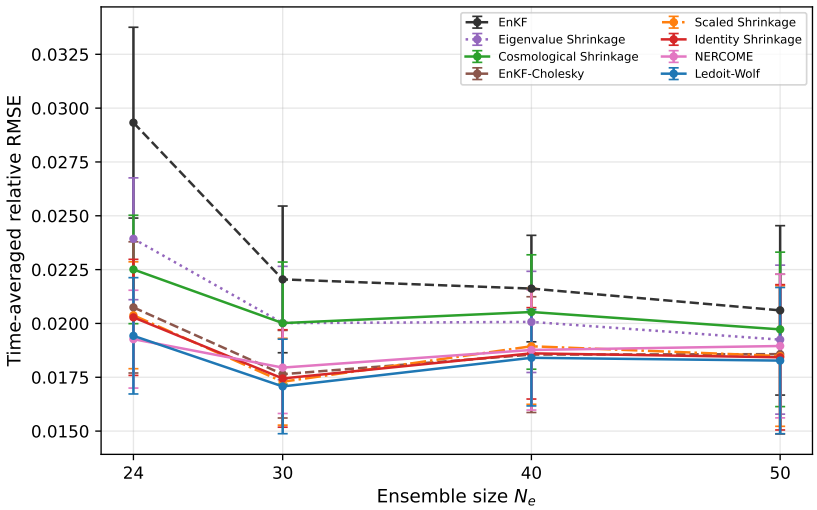

Fig. 5. Time-averaged relative RMSE vs. ensemble size for the AR(1) cosmological model. All methods show stable, monotonic behavior—unlike the Lorenz 96 case (Figure 2). Ledoit-Wolf achieves the lowest RMSE throughout. Error bars show ±1 standard deviation over 10 runs.

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